# Datasets Used in the Study

All datasets are public and de-identified. TCGA-UVM was used to train every model. The two GEO cohorts were used for external validation, and all three cohorts together formed the pooled and leave-one-cohort-out designs.

## 1. Summary

| Dataset | Role in this study | Platform | Samples used | Events | Outcome used |
|---|---|---|---|---|---|
| TCGA-UVM PanCancer Atlas | Training (all models); source of copy-number and methylation data for the anchoring prior; one of three cohorts in pooled and leave-one-cohort-out validation | RNA-seq (Illumina HiSeq, RSEM) + copy number + DNA methylation | 79 patients | 29 | Progression-free interval |
| GSE22138 | External validation (fixed-model evaluation); pooled and leave-one-cohort-out validation; monosomy-3-adjusted analysis | Affymetrix HG-U133 Plus 2.0 (GPL570) | 63 tumours (55 with chromosome 3 status) | 35 | Metastasis-free survival |
| GSE44295 | External validation (fixed-model evaluation); pooled and leave-one-cohort-out validation | Illumina HumanRef-8 v3 (GPL6883) | 57 primary tumours (of 63 profiles) | 24 | Time to metastatic death |

All analyses that combine TCGA with the GEO cohorts use the 9,067 genes measured on all three platforms. Each cohort was z-scored per gene within that cohort.

## 2. Training cohort: TCGA-UVM PanCancer Atlas

Role: training data for all models, with progression-free interval as the endpoint. Its copy-number and methylation profiles were used only to compute the anchoring score (the gene prior), never as inputs to the signature. Arm-level copy-number calls (chromosome 3 loss, 8q gain) and BAP1 mutation status were used only in the molecular-marker comparator. It was also one of the three cohorts in the pooled cross-validation and in leave-one-cohort-out validation.

- **Study page (cBioPortal):** https://www.cbioportal.org/study/summary?id=uvm_tcga_pan_can_atlas_2018
- **Direct download (all files, .tar.gz):** https://cbioportal-datahub.s3.amazonaws.com/uvm_tcga_pan_can_atlas_2018.tar.gz
- **Original data portal (GDC):** https://portal.gdc.cancer.gov/projects/TCGA-UVM

Files used from the archive: data_mrna_seq_v2_rsem.txt (expression), data_cna.txt (copy number), data_methylation_hm27_hm450_merged.txt (methylation), data_clinical_patient.txt (survival), data_armlevel_cna.txt (chromosome 3 / 8q status), data_mutations.txt (BAP1).

## 3. External validation cohort: GSE22138

Role: external validation of models trained once on TCGA and applied unchanged (fixed-model external evaluation); one of the cohorts in pooled and leave-one-cohort-out validation; the only external cohort with chromosome 3 status (55 of 63 patients), used for the monosomy-3-adjusted analysis. Original study: Laurent et al., Cancer Res 2011.

- **GEO series page:** https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE22138

- **Series matrix (file used):** https://ftp.ncbi.nlm.nih.gov/geo/series/GSE22nnn/GSE22138/matrix/ (GSE22138_series_matrix.txt.gz)
- **Platform annotation GPL570:** https://ftp.ncbi.nlm.nih.gov/geo/platforms/GPLnnn/GPL570/annot/GPL570.annot.gz (probe-to-gene mapping)

## 4. External validation cohort: GSE44295

Role: external validation of models trained once on TCGA and applied unchanged (fixed-model external evaluation); one of the cohorts in pooled and leave-one-cohort-out validation. Only the 57 primary tumours with known outcome were used; cell lines, normal melanocytes and stem cells were excluded. The outcome was time to metastatic death (patients alive, or dead without metastasis, were censored); the time of metastasis itself is not available in GEO, so this cohort is treated as a supporting cohort. No associated publication is listed in GEO.

- **GEO series page:** https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE44295
- **Series matrix (file used):** https://ftp.ncbi.nlm.nih.gov/geo/series/GSE44nnn/GSE44295/matrix/ (GSE44295_series_matrix.txt.gz)
- **Platform annotation GPL6883:** https://ftp.ncbi.nlm.nih.gov/geo/platforms/GPL6nnn/GPL6883/annot/GPL6883.annot.gz (probe-to-gene mapping)